

TwinWorld 2026 Challenge Final Verification Factsheet

Sparse-View 2D Gaussian Splatting with Corrected Depth Readout, and Optimal Covering Lattices for a Precision-Free Semantic Metric

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1. Team details

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- **Team website:** None.
- **Codabench user name:** kylechen1211, identical to the team name for unambiguous identification.
- **Codabench ID of the final submission:** **909861**, final phase, submitted 2026-08-31 09:56 UTC, archive `submission_v0.30_bcc019.zip`.
- **Final challenge score: 1.534.** Peak signal-to-noise ratio (PSNR) 19.95, structural similarity index measure (SSIM) 0.723, learned perceptual image patch similarity (LPIPS) 0.290, F-score 0.582, mean intersection over union in 3D (mIoU 3D) 0.341; equivalently `rendering_score` 0.610, `geometry_score` 0.582, `semantic_score` 0.341.
- **Source code, checkpoints and factsheet:** <https://github.com/kylevirtuous1211/ECCV-TwinWorld-challenge>
- **Complete results in the native representation:** <https://drive.google.com/drive/folders/1423PRQBWKVPb3fDJeUy4nV5z5sJk1Mf>

2. Method details

2.1. Overall pipeline

Fig. 1 summarizes our pipeline. We build on 2D Gaussian Splatting (2DGS) [4], implemented with `gsplat` [12]. The submission pipeline introduces no additional network, external data, or pretrained weights. Most of the performance gain comes from three modifications, each evaluated on the six scenes with released ground truth before inclusion in the final system.

1. A dense multi-view stereo (MVS) seed. The released sparse reconstruction covers only 0.07–0.20% of image pixels

and appears to be cropped from a larger model: all reprojection errors are zero, and the image identifiers fall in the 700–800 range although the model contains at least 819 images. We replace this initialization: we densely retriangulate the released images with the released poses held fixed, then run COLMAP [10] dense MVS. This change improves F-score by 0.10 on the held-out scenes.

2. Exposing and correcting the 2DGS depth rule. Because 2DGS represents surfaces as oriented discs, depth varies across each projected footprint and must be evaluated at the ray–disc intersection. The native `gsplat` path instead passes center depth as a per-Gaussian color channel, assigning a constant depth across the footprint. We recover the intersection depth from the outputs an unmodified `gsplat` installation already returns, so no change to the CUDA kernels is needed. On identical checkpoints from four TUM scenes, this correction improves `geometry_score` from **0.3888 to 0.4717** and produces point clouds with 7–11% fewer points.

3. Averaging three rendering seeds. Rendering quality varies across random seeds, but selecting the best seed requires ground truth, which is unavailable for the final-phase scenes. We instead train three models per scene and average their outputs per pixel. This improves `rendering_score` by 0.026 over the single seed that would otherwise have been submitted, and by 0.021 over the *best* of the three seeds on the released scenes. The same selection-free procedure applies directly to the held-out scenes. Fig. 2 shows the result against the released reference images on three scenes, and what averaging does not fix.

2.2. Extra data

No. All inputs are drawn from the released `TwinWorld_Datasets` (Hugging Face snapshot 8d5dfff) or from the scores returned for our own submissions. We use no external ground truth, data outside the challenge release, or pretrained scene reconstructions. The only pretrained weights used anywhere in our experiments are those of the VGG network inside the reference LPIPS implementation.

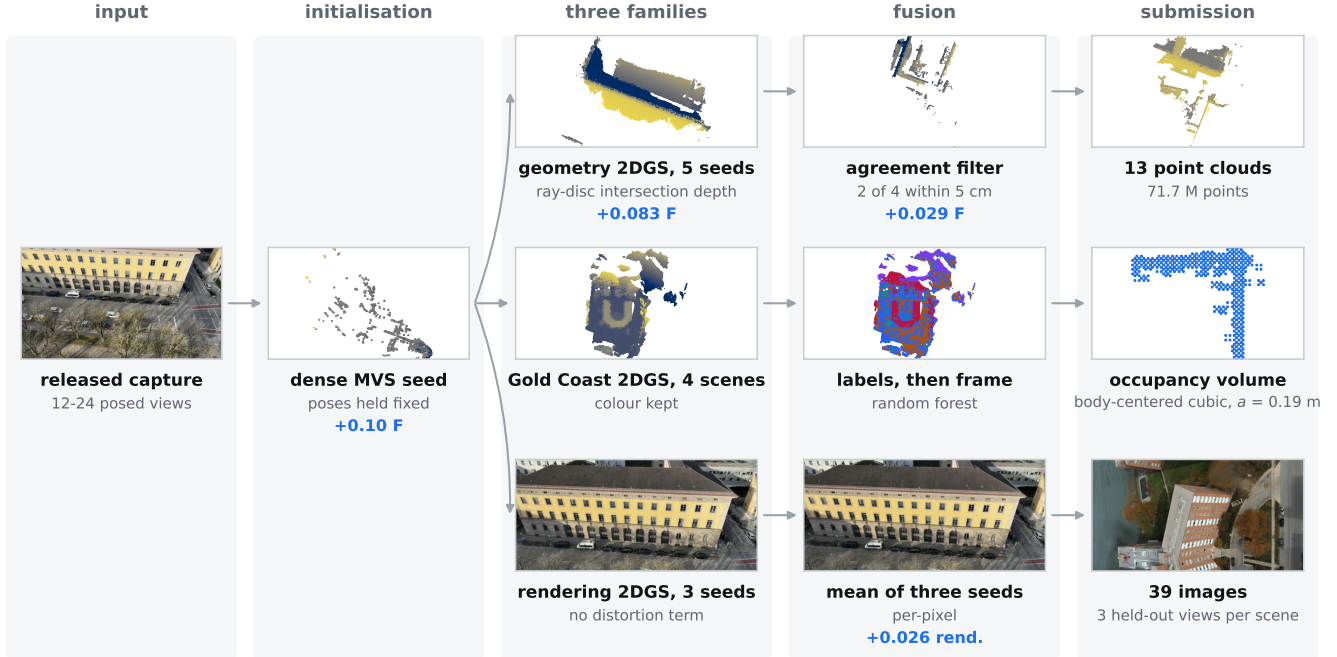


Figure 1. Overview of the pipeline. A common dense initialization is used to train three families of Gaussian models: TUM geometry (five seeds with agreement filtering), Gold Coast geometry (also providing color features to the semantic classifier), and rendering (three seeds with per-pixel averaging). The occupancy volume in the submission column is described in Sec. 4. Gains shown in blue are measured on the six scenes with released ground truth.

It is used only for local LPIPS evaluation, not in any submission.

We do use the ground truth the organisers released. The label classifier is trained on it: every training label is taken from the nearest point of the released ground-truth clouds of `scene_009` and `scene_010`, which is what those two scenes ship them for. Two further uses, in which development-set ground truth informs predictions on scenes that ship none, are summarized in Sec. 4.

2.3. Training strategy

We train separate models for geometry and rendering because the two objectives favor different configurations.

Geometry. Nine TUM scenes at five seeds each, and four Gold Coast scenes. 2DGS, 7000 iterations for TUM and 15,000 for Gold Coast, normal weight 0.05, distortion weight 0.01, intersection-depth readout, and initialization from the dense MVS seed. Seed 0 is fused into a point cloud, while seeds 1–4 are used only for agreement filtering. A point from seed 0 is retained if at least two of the other four seeds place a point within 5 cm. On the four TUM scenes with released ground truth, this filter improves `geometry_score` from 0.4730 to 0.5016 (+0.029). It removes 49% of points from the released-seed clouds on which it was tuned and 28.5% from the submitted dense-MVS clouds. This reduction allows the submission to retain

the full 2 cm fusion resolution while remaining within the evaluator’s point budget.

Rendering. Thirteen scenes at three seeds each. We train 2DGS for 7000 iterations with a normal weight of 0.05, a distortion weight of 0, and the same dense MVS initialization. The three outputs are averaged per pixel.

Semantics. We train a random forest [9] with 200 trees, a maximum depth of 32, and a minimum leaf size of 2. Its inputs comprise geometric features and rendered color. The classifier is fitted to reconstructed clouds rather than ground-truth clouds. In the corrected coordinate frame, training on the reconstruction and including color jointly improve `semantic_score` by 0.15.

Frame correction. The released Gold Coast ground truth is displaced by 1.1–1.4 m from the camera coordinate frame defined by its corresponding images. This discrepancy originates in the released data rather than in our reconstruction: registering the released sparse reconstruction directly to the released ground-truth cloud, with nothing of ours involved, recovers the same 0.25° rotation. We measure this discrepancy on the two Gold Coast scenes with released ground truth. As a control, the corresponding offsets on the four TUM scenes range from 0.014 to 0.074 m.

2.4. Complexity and cost

Table 1 gives the cost of one pass over the dataset.

Model family	Scenes	Gauss.	Training
Geometry, TUM, per seed	9	19.1 M	not logged
Geometry, Gold Coast	4	28.7 M	not logged
Rendering, per seed	13	25.5 M	43 min

Table 1. Cost of one pass over the dataset. Gaussians are summed over the scenes in the row. Training wall clock was recorded per model only for the rendering runs, whose receipts give 2.5 to 5.5 minutes each and 43 minutes for a seed over all 13 scenes. The geometry runs timed only their fusion step, 45 to 1623 seconds per model with a mean of 283 over 49 models, so their training time is a gap in our records rather than a small number.

TUM geometry models contain 1.4–2.5 M Gaussians per scene, and Gold Coast models 7.0–7.5 M. A rendering model trains in 2.5–5.5 minutes. Rebuilding all 13 scenes takes approximately six hours on a single RTX 6000 Ada, or two hours using a Slurm array on H200 nodes. A sampled geometry job used 7 GB of the available 48 GB of graphics processing unit (GPU) memory. We train no neural network beyond the Gaussian representation, so a conventional parameter count does not apply. The submitted archive contains 71.7 M points across 13 point clouds and 39 rendered images.

2.5. Inference and submission

No retraining is performed during submission generation. Geometry models are rendered to depth maps, fused, agreement-filtered, and thinned. Gold Coast models are assigned semantic labels and transformed to the corrected frame; `scene_011` and `scene_012` are additionally converted to the occupancy representation described in Sec. 4. Rendering outputs are averaged across three seeds. The results bundle carries the complete conversion procedure, command by command. Following it reproduces the two submitted Gold Coast clouds byte for byte, and all 39 renders in the results bundle are byte-identical to those in the submitted archive.

2.6. Experimental results

Our final score is 1.53, fourth among 21 entries in the closed final phase. Our F-score of 0.58 is close to the challenge median of 0.57, whereas our mIoU of 0.34 ranks fourth, compared with a median of 0.22. Our ranking is therefore driven more by semantic performance than by geometry. Table 2 gives the phase in full, and Fig. 3 shows what the geometry term looks like on a scene that ships ground truth: our cloud beside the truth, and the distance from each truth point to the nearest point we predict.

Table 3 places our two phases side by side. We led the development phase at 1.60 and finished fourth at 1.53. Rendering and geometry are essentially unchanged between the two; the whole difference is semantic, mIoU 0.52 against 0.34. **We do not read this as overfitting in the usual sense,**

because the two phases score disjoint scene sets and a 0.18 gap between scene sets is entirely possible on its own. What we can say is narrower and still useful: the Gold Coast semantic term is the least stable part of our submission across scenes, our development-phase measurements were the least transferable, and the frame offset for the two unlabelled test scenes – which we could only infer – is the most likely place for that instability to live.

Interpreting the geometry score. `geometry_score` averages F-scores at 5, 10, and 20 cm, which can obscure performance at the strictest threshold. On the four TUM scenes with released ground truth (F-score is evaluated only on TUM scenes), one development archive obtains **0.357 at 5 cm, 0.521 at 10 cm, and 0.637 at 20 cm**. Their mean is 0.505, substantially closer to the 20 cm result than to the 5 cm result. Because this archive predates the final cloud family, these values illustrate the composition of the metric rather than decompose the final score of 0.582. The leaderboard reports only the mean over the five evaluated scenes.

Scope of the depth ablation. The gain of 0.083 reported below predates the dense MVS initialization and should not be interpreted as the gain of the final system. On a submitted checkpoint from `scene_000`, seed 0, the correction increases F-score from 0.5949 to 0.6114; precision at 5 cm decreases from 0.447 to 0.408, while recall increases from 0.437 to 0.510. This comparison covers one scene rather than four, and we have not isolated the source of the difference. We therefore attribute 0.083 only to the earlier model family on which it was measured.

The principal internal ablations, evaluated on scenes with released ground truth, are summarized below:

Change	Effect
center → intersection depth	+0.083 F
sparse → dense MVS init.	+0.10 F (LB)
one seed → 5-seed vote	+0.029 F
single seed → 3-seed mean	+0.026 rend.
surface → labeled volume	+0.06 mIoU (LB)

Here, “LB” denotes an effect measured through leaderboard submissions.

2.7. Novelty and relation to prior work

Our depth component does not change the 2DGS formulation. It exposes an implicit requirement of the rendering geometry, provides a patch-free correction for the gsplat readout, and quantifies its effect on the TwinWorld benchmark. Without the correction, the native path measures center-depth surrogates rather than the surface depth defined by 2DGS [4]; the discrepancy also propagates to gsplat’s normal-consistency loss.

The lattice construction in Sec. 4 follows Bambah [2], as presented by Conway and Sloane [3]. Alam and Haas [1]

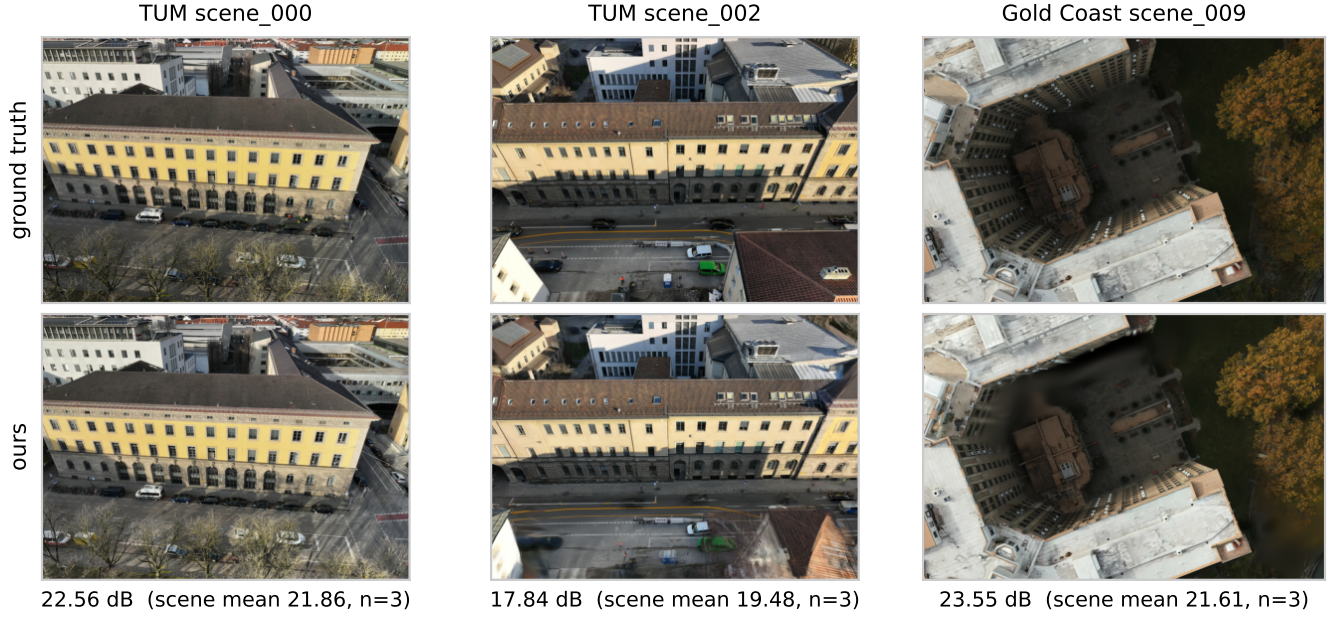


Figure 2. Qualitative rendering results on held-out views. The top row shows the released reference images, and the bottom row shows our submitted renders, obtained by averaging three seeds. Each scene contains three held-out views; we show the view with median PSNR and report the scene-level mean. Averaging reduces seed-dependent speckle but does not resolve structural errors: thin foreground objects and sparsely observed surfaces remain blurred across all seeds. We do not decompose the rendering metric by error type, so this is qualitative.

#	team	Final	PSNR	SSIM	LPIPS	F-score	mIoU 3D
1	totototo	1.67	21.23	0.75	0.28	0.56	0.47
2	lionelsy	1.61	20.03	0.7	0.29	0.64	0.37
3	windrise	1.6	19.75	0.72	0.28	0.61	0.38
4	kylechen1211	1.53	19.95	0.72	0.29	0.58	0.34
5	manish	1.48	20.9	0.74	0.3	0.64	0.22
6	qqqqyyyyzzzz	1.41	19.11	0.63	0.35	0.62	0.24
7	qqyyyyzzzz	1.41	19.09	0.63	0.35	0.62	0.24
8	xcyin	1.41	20.07	0.69	0.31	0.62	0.19
9	tos111	1.4	21.23	0.75	0.28	0.56	0.2
10	qiuyizhuo	1.38	18.91	0.63	0.36	0.62	0.21
11	qinwenbin	1.36	18.54	0.55	0.35	0.57	0.26
12	qwbqwbqwb	1.34	18.54	0.55	0.35	0.57	0.25
13	yruchen	1.3	19.86	0.67	0.4	0.6	0.15
14	toscanago	1.3	21.23	0.75	0.28	0.53	0.13
15	acceon	1.29	19.22	0.58	0.39	0.46	0.3
16	fxfyhx	1.27	17.37	0.49	0.44	0.59	0.22
17	hclab	1.18	18.58	0.66	0.38	0.56	0.07
18	vapaus	1.17	17.65	0.51	0.41	0.57	0.12
19	zhayi	1.11	18.17	0.65	0.37	0.48	0.09
20	licheng	1.09	17.92	0.63	0.39	0.49	0.07
21	ylnnng	1.06	17.0	0.47	0.4	0.51	0.08
median of 21		1.36	19.11	0.65	0.35	0.57	0.22

Table 2. The closed final phase in full, all 21 entries, read from the public endpoint https://www.codabench.org/api/phases/29458/get_leaderboard/ and kept verbatim beside the factsheet sources. Our row is shaded.

previously applied optimal covering lattices to sensor coverage and derived the same constant of 1.859. SparseOcc [7]

also showed that thickening a surface can exploit a one-sided occupancy metric, reporting improvements of 5–15 mIoU.

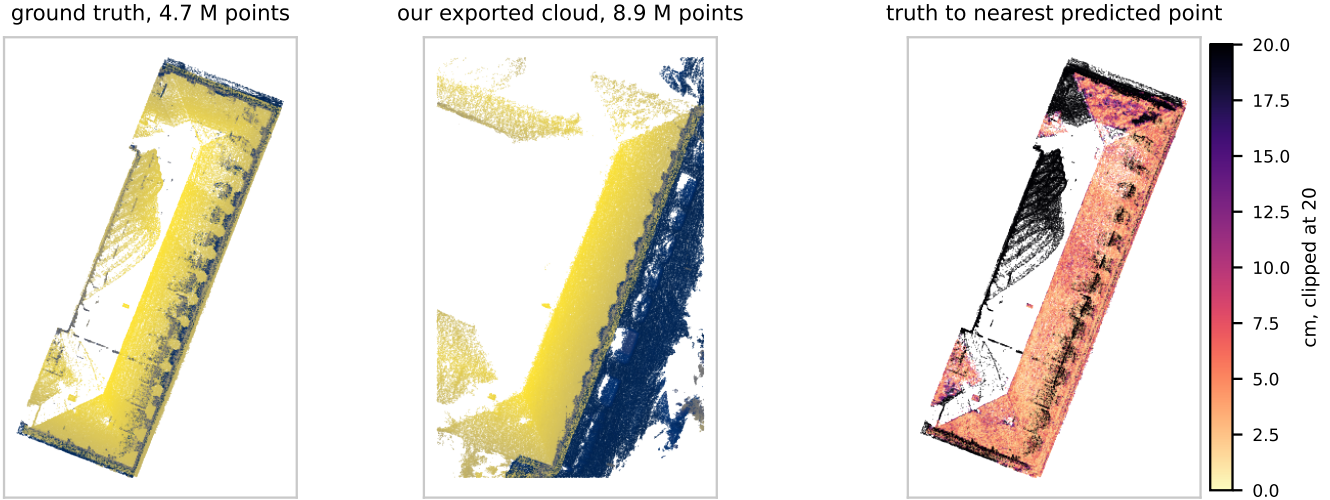


Figure 3. The geometry term on `tum/scene_000`, viewed from above. Left: released ground truth. Middle: the cloud our pipeline exports. Right: the distance from each ground-truth point to the nearest point we predict, which is the recall side of the F-score. Both clouds are cropped to the truth’s bounding box and voxel-downsampled at 2 cm first, exactly as the scorer does. Roof surfaces sit within a few centimetres; the dark regions are courtyard and vegetation seen from too few views. This exported cloud scores $F = 0.611$; the copy inside the submitted archive scores 0.413, because the archive spends its point budget on the seven scenes the final phase scores, thinning the six it does not: `tum/scene_000` to `scene_003` and `gold_coast/scene_009` and `scene_010`.

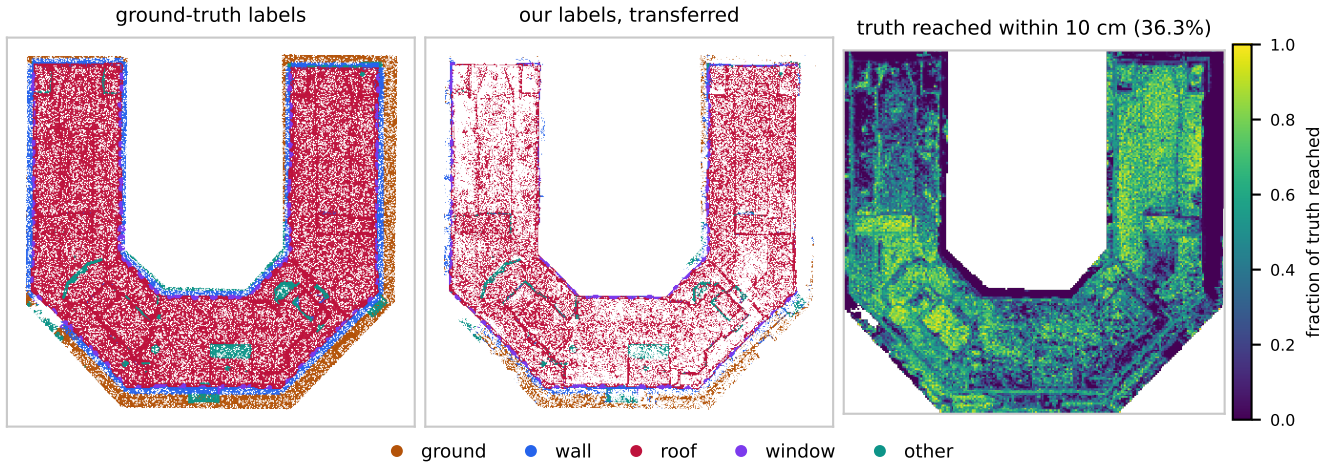


Figure 4. The semantic term on `gold_coast/scene_009`, viewed from above. Left: released ground-truth labels. Middle: our submitted labels, transferred to the truth exactly as the metric does. Right: the fraction of ground-truth points in each cell that have any prediction within 10 cm. Per-class IoU is 0.220 ground, 0.293 wall, 0.456 roof, 0.244 window and 0.296 other, a mean of 0.302. The third panel is the point of Sec. 5.2: **63.7% of the ground truth has nothing of ours within 10 cm at all**, and every one of those points is an error the metric charges us.

Neither the specific system nor the benchmark analysis presented in this submission has been published previously.

3. Ensemble and fusion strategies

The final submission uses two ensemble strategies.

Five-seed agreement filter (geometry). We train five geometry models per TUM scene, differing only in random seed. The point cloud fused from seed 0 is filtered

by retaining points supported within 5 cm by at least two of the remaining four seeds. This operation is an agreement filter rather than a fusion step, because it adds no points. On the four scenes with released ground truth, it improves `geometry_score` by 0.029 and removes 28.5% of points from the submitted clouds.

Three-seed render averaging. We compute the per-pixel mean of three rendering models. This improves

phase	rank	Final	PSNR	F	mIoU
development	1 of 19	1.60	19.68	0.51	0.52
final	4 of 21	1.53	19.95	0.58	0.34

Table 3. Our best entry in each phase, same source as Tab. 2 (development is phase 29457; submission 894488 and 909861). We led the

rendering_score by 0.026 and outperforms the best individual seed on five of the six measurable scenes. Pixel-wise median aggregation is consistently worse than the mean (0.5979 versus 0.6026), which fits approximately symmetric seed-to-seed errors.

We also evaluated, but did not submit, a distance-gated floater-pruning strategy. Although it improved three scenes, its mean effect over 18 paired comparisons across six scenes and three seeds was -0.0012 .

4. Disclosed properties of the submission

Three properties of the final submission.

1. Released development-set ground truth is used to align test point clouds. Because `scene_009` and `scene_010` include ground truth, their frame offsets can be measured directly. Ground truth is unavailable for `scene_011` and `scene_012`, two of the seven scenes evaluated in the final phase, so their offsets are inferred. We transfer the offsets measured on the first two scenes and also transform the ground-truth cloud of `scene_010` into the camera frame of `scene_011` through a joint COLMAP reconstruction of the three registrable, partially overlapping Gold Coast scenes. Two of our submissions isolate the effect: `v0.19` and `v0.20` differ only in whether this offset is applied to those two scenes, and scored 1.2010 and 1.27, with PSNR, SSIM, LPIPS and F-score unchanged and the whole difference in mIoU, 0.1207 to 0.19. We read “external ground-truth data” in the challenge terms as data from outside the release, which this is not; if the intended reading is stricter, this is the item it applies to. The offset is a single parameter of the submission-packing step, and reverting it changes one line.

2. One of the three offset estimates uses scores returned for our own submissions. We model pooled mIoU as $ag(o - t)$, where g is a response curve measured on the two scenes with released ground truth, and fit t using five of our scored submissions.

3. For `scene_011` and `scene_012`, the submitted point cloud is a labeled occupancy volume rather than a labeled surface. This representation optimizes the stated semantic metric but does not improve reconstruction quality. It contributes more to our semantic score than any other individual component. Fig. 5 visualizes the representation, and Sec. 5.2 analyzes the underlying metric behavior independently of our entry.

5. Other details

5.1. Impressions and practical challenges

The dataset is well suited to evaluating digital-twin reconstruction, and the three-part score prevents appearance alone from determining the ranking. The main practical challenge was model selection under limited supervision: the final phase evaluates seven scenes without released ground truth, so choices validated on the six development scenes have to transfer to unseen data. This led to two notable errors. First, we selected a Gold Coast representation using development clouds thinned to 12 cm, whereas the target clouds used 6 cm spacing; an improvement factor we expected to be 2.26 was only 1.29 on the leaderboard. Second, one scene was submitted with approximately one quarter of its spatial extent missing, and a ground-truth-free diagnostic identified the issue only after three submissions.

A second difficulty concerns the undocumented point-count limit. Archives above this limit are rejected with a null exit code and no diagnostic message. We used three submissions to estimate the threshold. Publishing the limit and returning an explicit error message would make this failure mode substantially easier to diagnose.

5.2. Comments on the evaluation metric

We identify one property of the semantic score that may merit attention in future editions.

The Gold Coast semantic metric performs a ground-truth-driven nearest-neighbor lookup within 10 cm and contains no precision term: a predicted point with no nearby ground-truth point incurs no penalty. On the two scenes with released ground truth, removing every false positive from our submission increases pooled mIoU from 0.3104 to 0.3207. In contrast, matching every ground-truth point while retaining our assigned labels would increase mIoU to 0.94. Fig. 4 shows both halves of this on one scene: our labels beside the truth, and how much of the truth we are near enough to label at all. Under this analysis, the metric is approximately 60 times more sensitive to coverage than to precision.

Under this scoring rule, a labeled occupancy volume can be preferable to a surface representation. The optimization then becomes a covering problem: place N points to maximize the proportion of ground-truth points within 10 cm of a prediction. We use a body-centered cubic lattice, the lowest-density *lattice* covering of three-dimensional space [2, 3]. It guarantees a 10 cm match using 349 points per cubic meter, compared with 649 for a simple cubic grid.

Fig. 6 is the sweep behind that choice. Of the 29 constructions run, the 19 whose lattice has a defined covering radius are plotted, scored on the two Gold Coast scenes that ship ground truth at the residual frame error the offset correction is expected to leave, with the selection criterion fixed

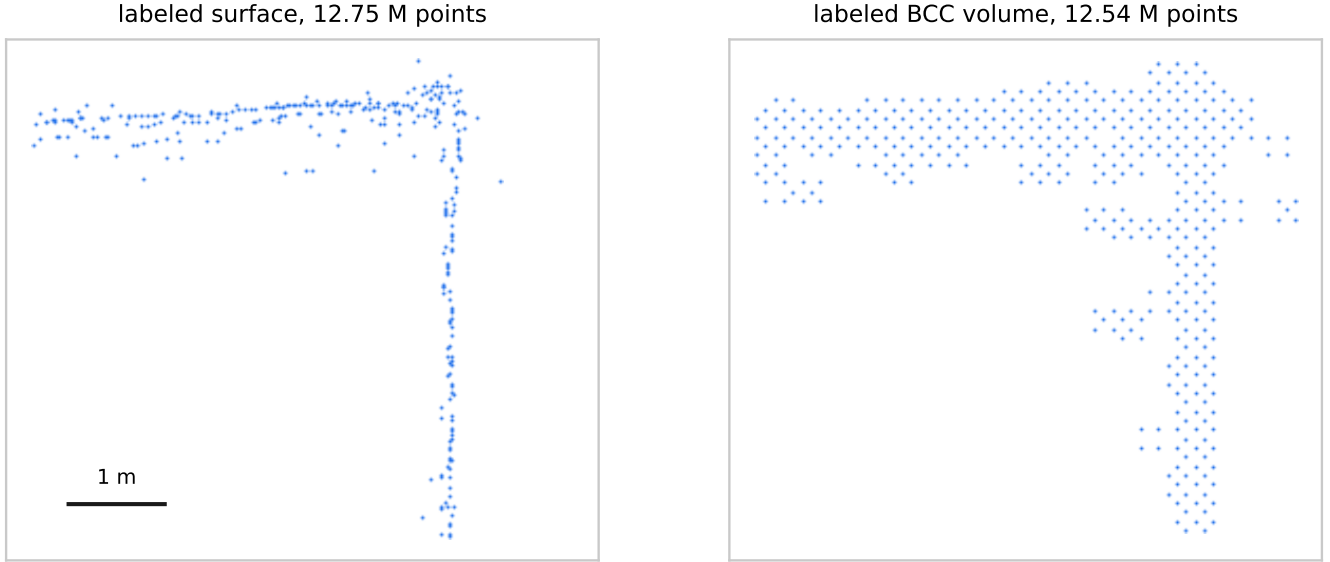


Figure 5. Visualization of disclosed property 3 using the submitted archives. We show corresponding 8×5 m vertical slices of `scene_011`, each 12 cm thick. Left: the labeled surface used through submission v0.27. Right: the body-centered cubic (BCC) occupancy volume in the final submission, using lattice constant $a = 0.19$ m within a 0.17 m shell. The point counts are nearly equal; the representations differ primarily in whether points lie on the surface or fill its neighborhood. This change improves leaderboard mIoU by 0.06, despite producing a less faithful geometric representation; see Sec. 5.2.

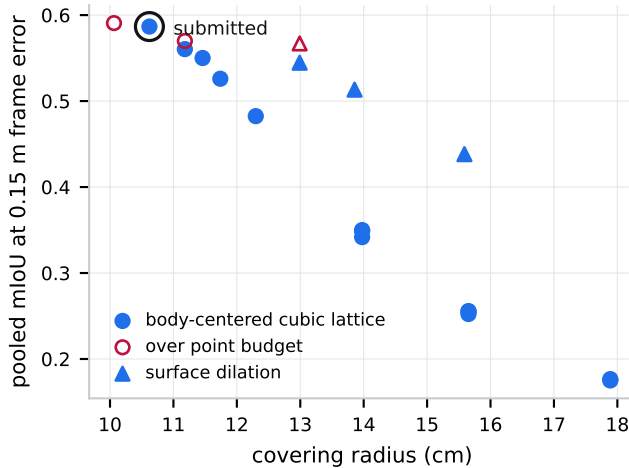


Figure 6. The lattice sweep. Each point is one construction of the Gold Coast cloud, scored on the two scenes that ship ground truth. Filled markers are inside the per-cloud point budget that has already been accepted by the evaluator; open red markers exceed it. Within the lattice family the score tracks the covering radius and is nearly indifferent to shell thickness, which is why the submitted configuration is the smallest admissible radius rather than the thickest shell. The surface-dilation variants are a different construction and are not comparable to the lattices at equal radius; see the text.

before the grid ran. Two things in it are worth separating.

Within the lattice family the score is governed by the covering radius and almost nothing else. It falls monotonically

from 0.587 at a radius of 10.6 cm to 0.177 at 17.9 cm, while the covered volume stays within $36,000\text{--}51,000\text{ m}^3$. Shell thickness barely registers: at a radius of 15.7 cm, tripling the shell from 0.45 to 0.75 m moves the score from 0.2523 to 0.2556, and the three shells tried at 14.0 cm all land within 0.008 of each other. We had expected a coarser lattice to buy back its loss through thickness, and it does not.

The comparison does not carry across constructions, and Fig. 6 separates them for that reason. The surface-dilation variants, which thicken the reconstructed surface instead of filling space, sit above the lattice trend at equal covering radius: a dilation at 13.0 cm scores 0.545 where a lattice at 12.3 cm scores 0.483. They place points only where the surface already is, which is a real advantage that covering radius alone does not express. What the lattice buys is the guarantee, not a uniformly better score at every budget.

The covered volume remains within $36,000\text{--}51,000\text{ m}^3$, while the score varies by a factor of three. This indicates that point placement, rather than total covered volume, governs the metric. An oracle with access to ground truth matches the lattice score using 17 times fewer points, suggesting that the point budget is not the limiting factor; approximately 30% of ground-truth points are not close to any reconstructed surface.

Comparison with related benchmarks. KITTI-360 uses a similar ground-truth-driven nearest-neighbor label-transfer metric for semantic mapping [6]. However, it also reports geometric accuracy for the same predicted points,

so dilating a point cloud degrades its geometric score. ScanNet-family benchmarks evaluate predictions on a provided point set; consequently, predictions correspond to ground-truth points and false positives are counted directly. In TwinWorld, `geometry_score` is evaluated on TUM scenes and `semantic_score` on a disjoint set of Gold Coast scenes. The combined score therefore does not penalize volumetric predictions on the scenes used for semantic evaluation. We recommend addressing this cross-scene decoupling rather than changing the mIoU definition itself.

5.3. Suggestions for future editions

1. **Add a precision component to semantic evaluation and evaluate it on the same scenes as geometry.** This follows the surface evaluation of Tanks and Temples [5] and the paired semantic and geometric reporting of KITTI-360 [6], and would remove the incentive for volumetric predictions.
2. **Publish the point-count limit and report an explicit error when it is exceeded.** This would prevent teams from spending submissions solely to diagnose an undocumented constraint.
3. **Reassess the reliability of the 5 cm threshold.** The TUM2TWIN data-quality analysis [11, Tab. 2] reports absolute and relative accuracies of 0.08 m and 0.05 m, respectively, for the uncrewed aerial system (UAS) laser scan. UseGeo [8] reports a mean cloud-to-cloud error of 8.5 cm for conventional dense photogrammetry relative to an airborne light detection and ranging (LiDAR) survey, using 224 images and at least eight rays per object point. With only 12–24 views, F-score at 5 cm may partly reflect uncertainty in the reference data. We present this as a question because we have not directly measured reference accuracy.
4. **Release ground truth for one withheld scene after the challenge closes.** This would enable teams to assess how design choices transfer from the six development scenes to the seven held-out scenes.
5. **State whether development-set ground truth may inform test-set predictions.** The Gold Coast georeferencing offset makes this distinction consequential, but the current terms do not resolve it explicitly.

5.4. Limitations of the released checkpoints

The 39 rendering runs were executed without checkpoint saving; consequently, the Gaussian parameters that produced the submitted images are unavailable. We tested whether deterministic retraining could recover them. Two runs using identical commands, random seeds, and GPUs differed by 31,224 Gaussians at termination; none of their three rendered frames were byte-identical, and the PSNR between corresponding runs ranged from 15.7 to 23.8 dB. The training code seeds only the framework’s global ran-

dom number generator, while the gsplat backward pass accumulates gradients atomically, in an order that is not fixed between launches.

The results bundle therefore contains every rendered image from all three seeds, the 39 averaged images that are byte-identical to the submitted archive, and the complete configuration and metrics for each run. We do not provide retrained weights as substitutes for the original checkpoints. Checkpoints are available for every submitted point cloud except seed 0 of `tum/scene_006`, which was also trained without checkpoint saving. Its remaining four agreement-filtering seeds, exported point cloud, color data, and generation receipt are included.

Acknowledgements

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